

Kummer-Fourier Smoothing and the Random-Walk Bridge

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2026-05-13

Abstract

Paper 4 isolated a random-walk smoothing bridge (KRWS) as a named input in the conditional Erdős–Graham #203 route. Papers 8–11 now make that bridge finite and exact for the Bernoulli-Kummer obstruction law actually used by the CRT sieve model. This note proves the resulting smoothing theorem: weighted prime Hecke non-concentration gives exponential L^2 and total-variation flattening of the Bernoulli obstruction distribution, with collision probability $\ell^{-r} + O_A((\log Q)^{-A})$ in the high- ℓ range once PHNC is supplied. It also proves that the analogous deterministic prime-vector walk is not controlled by linear hyperplane non-concentration: affine hyperplane concentration gives a zero spectral gap. Thus KRWS splits into a solved Bernoulli branch and a deterministic branch requiring affine non-concentration. The result does not prove PHNC, Kummer-BV, BVK, or Erdős–Graham #203.

1 Finite Fourier conventions

Let

$$G = \mathbb{F}_\ell^r, \quad |G| = \ell^r,$$

and let u denote the uniform probability measure on G . For a probability measure μ on G , define

$$\widehat{\mu}(A) = \sum_{x \in G} \mu(x) e_\ell(A \cdot x), \quad A \in G,$$

where $e_\ell(t) = \exp(2\pi i t / \ell)$. Then

$$\|\mu - u\|_2^2 = \ell^{-r} \sum_{A \neq 0} |\widehat{\mu}(A)|^2,$$

and

$$\|\mu - u\|_{\text{TV}} \leq \frac{1}{2} \ell^{r/2} \|\mu - u\|_2.$$

The collision probability of μ is

$$\text{Col}(\mu) = \sum_{x \in G} \mu(x)^2 = \ell^{-r} + \|\mu - u\|_2^2.$$

2 The deterministic affine trap

The first point is negative and structural. Linear hyperplane non-concentration does not imply smoothing for the deterministic step measure

$$\mu_Q = \frac{1}{|Q|} \sum_{q \in Q} \delta_{x_q}.$$

Proposition 1 (Affine hyperplane trap). *Let $r \geq 2$, let $A_0 \in \mathbb{F}_\ell^r \setminus \{0\}$, and let μ be the uniform measure on the affine hyperplane*

$$H_{A_0,1} = \{x \in \mathbb{F}_\ell^r : A_0 \cdot x = 1\}.$$

Then μ assigns mass at most $1/\ell$ to every linear hyperplane $\{x : A \cdot x = 0\}$, but

$$|\widehat{\mu}(A_0)| = 1.$$

*Consequently μ^{*n} does not converge to u for any sequence $n \rightarrow \infty$.*

Proof. If A is a nonzero scalar multiple of A_0 , then $\{A \cdot x = 0\} \cap H_{A_0,1} = \emptyset$. If A is not a scalar multiple of A_0 , then the two independent affine-linear equations cut $H_{A_0,1}$ in codimension one inside $H_{A_0,1}$, hence in relative mass $1/\ell$. This proves the claimed linear hyperplane non-concentration.

For $x \in H_{A_0,1}$, the character $e_\ell(A_0 \cdot x)$ is the constant $e_\ell(1)$. Thus $\widehat{\mu}(A_0) = e_\ell(1)$, so $|\widehat{\mu}(A_0)| = 1$. Since $\widehat{\mu^{*n}}(A_0) = \widehat{\mu}(A_0)^n$, the same coefficient has modulus one for every n , whereas the uniform measure has zero nontrivial Fourier coefficients. $\square$

3 Deterministic smoothing under affine non-concentration

The correct deterministic finite theorem uses affine hyperplanes, not only linear kernels.

Theorem 1 (Affine non-concentration smoothing). *Let μ be a probability measure on $G = \mathbb{F}_\ell^r$. Assume that for some $0 < \delta \leq 1/2$,*

$$\max_{A \neq 0, t \in \mathbb{F}_\ell} \mu(\{x : A \cdot x = t\}) \leq 1 - \delta.$$

Then for every $A \neq 0$,

$$|\widehat{\mu}(A)|^2 \leq 1 - 2\delta(1 - \delta) \left(1 - \cos \frac{2\pi}{\ell}\right).$$

Consequently, for every $n \geq 1$,

$$\|\mu^{*n} - u\|_2^2 \leq (1 - \ell^{-r}) \exp \left(-2n\delta(1 - \delta) \left(1 - \cos \frac{2\pi}{\ell}\right) \right),$$

and

$$\|\mu^{*n} - u\|_{\text{TV}} \leq \frac{1}{2}(\ell^r - 1)^{1/2} \exp \left(-n\delta(1 - \delta) \left(1 - \cos \frac{2\pi}{\ell}\right) \right).$$

Proof. Fix $A \neq 0$, and let $Y = A \cdot X \in \mathbb{F}_\ell$, where X has law μ . Let $\rho(t) = \mathbb{P}(Y = t)$. The affine non-concentration assumption gives $\max_t \rho(t) \leq 1 - \delta$. Since $0 < \delta \leq 1/2$,

$$\sum_t \rho(t)^2 \leq (1 - \delta)^2 + \delta^2.$$

Let Y' be an independent copy of Y . Then

$$\mathbb{P}(Y \neq Y') = 1 - \sum_t \rho(t)^2 \geq 2\delta(1 - \delta).$$

Also

$$|\widehat{\mu}(A)|^2 = \mathbb{E} e_\ell(Y - Y').$$

The imaginary part vanishes by symmetry, and when $Y - Y' \neq 0$ the real part is at most $\cos(2\pi/\ell)$. Hence

$$|\widehat{\mu}(A)|^2 \leq 1 - \mathbb{P}(Y \neq Y') \left(1 - \cos \frac{2\pi}{\ell}\right),$$

which proves the spectral gap. Plancherel applied to μ^{*n} , together with

$$(1 - x)^n \leq e^{-nx},$$

gives the L^2 bound. The total-variation estimate follows from Cauchy–Schwarz:

$$\|\mu^{*n} - u\|_{\text{TV}} \leq \frac{1}{2} \ell^{r/2} \|\mu^{*n} - u\|_2.$$

□

Corollary 1 (Deterministic KRWS from affine PHNC). *For $r = 2$, if $\ell \leq (\log Q)^B$, $\delta = \ell^{-C}$, and*

$$n\delta(1 - \delta) \left(1 - \cos \frac{2\pi}{\ell}\right) \geq A \log \log Q + \log \ell + O(1),$$

then

$$\text{Col}(\mu^{*n}) = \ell^{-2} + O_A((\log Q)^{-A}).$$

Using $1 - \cos(2\pi/\ell) \geq 8/\ell^2$, it is enough to take

$$n \gg_{A,B,C} \ell^{C+2} \log \log Q.$$

4 The Bernoulli-Kummer obstruction walk

The deterministic affine trap is not the obstruction law used in the CRT model. The actual finite object is the Bernoulli-Kummer obstruction walk

$$\nu = *_{q \in \mathcal{Q}} ((1 - p_q)\delta_0 + p_q\delta_{x_q}),$$

the law of

$$Y = \sum_{q \in \mathcal{Q}} B_q x_q,$$

where the B_q are independent Bernoulli variables with parameters p_q . The independence is the CRT independence used in Paper 8.

Definition 1 (Weighted PHNC mass). *Define*

$$W_\ell(\mathcal{Q}) = \min_{A \neq 0} \sum_{q: A \cdot x_q \neq 0} p_q(1 - p_q).$$

Theorem 2 (Bernoulli-Kummer smoothing). *For the Bernoulli-Kummer obstruction law ν ,*

$$\|\nu - u\|_2^2 \leq (1 - \ell^{-r}) \exp \left(-\frac{16}{\ell^2} W_\ell(\mathcal{Q}) \right),$$

and

$$\|\nu - u\|_{\text{TV}} \leq \frac{1}{2} (\ell^r - 1)^{1/2} \exp \left(-\frac{8}{\ell^2} W_\ell(\mathcal{Q}) \right).$$

Equivalently,

$$\text{Col}(\nu) = \ell^{-r} + O \left(\exp \left(-\frac{16}{\ell^2} W_\ell(\mathcal{Q}) \right) \right),$$

with the explicit error factor bounded by $1 - \ell^{-r}$.

Proof. For $A \neq 0$, independence gives

$$\widehat{\nu}(A) = \prod_{q \in \mathcal{Q}} ((1 - p_q) + p_q e_{\ell}(A \cdot x_q)).$$

Therefore

$$|\widehat{\nu}(A)|^2 = \prod_{q \in \mathcal{Q}} \left[1 - 2p_q(1 - p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right) \right],$$

which is the Paper 8 squared-modulus identity after normalization by $\varphi(P)^2$. Using $1 - y \leq e^{-y}$ and the universal gap

$$1 - \cos \frac{2\pi t}{\ell} \geq \frac{8}{\ell^2} \quad (t \neq 0),$$

we obtain

$$|\widehat{\nu}(A)|^2 \leq \exp \left(-\frac{16}{\ell^2} \sum_{q: A \cdot x_q \neq 0} p_q(1 - p_q) \right) \leq \exp \left(-\frac{16}{\ell^2} W_{\ell}(\mathcal{Q}) \right).$$

Plancherel over the $\ell^r - 1$ nontrivial characters gives

$$\|\nu - u\|_2^2 = \ell^{-r} \sum_{A \neq 0} |\widehat{\nu}(A)|^2 \leq (1 - \ell^{-r}) \exp \left(-\frac{16}{\ell^2} W_{\ell}(\mathcal{Q}) \right).$$

The total-variation and collision estimates follow from the identities in the opening section. $\square$

Corollary 2 (PHNC high- ℓ consequence). *Assume $0 < \alpha \leq p_q \leq 1 - \alpha$ and PHNC in the form*

$$\#\{q \in \mathcal{Q} : A \cdot x_q \neq 0\} \geq \ell^{-C} |\mathcal{Q}| \quad (A \neq 0).$$

Then

$$W_{\ell}(\mathcal{Q}) \geq \alpha(1 - \alpha)\ell^{-C} |\mathcal{Q}|,$$

and therefore

$$\|\nu - u\|_2^2 \leq (1 - \ell^{-r}) \exp \left(-16\alpha(1 - \alpha)\ell^{-C-2} |\mathcal{Q}| \right).$$

If $\ell \leq (\log Q)^B$ and $|\mathcal{Q}| \asymp Q/(\ell \log Q)$, the exponent is

$$\gg_{\alpha, B, C} \frac{Q}{\ell^{C+3} \log Q} \geq \frac{Q}{(\log Q)^{B(C+3)+1}}.$$

Thus for fixed $A_0 > 0$ and fixed r, B, C ,

$$\text{Col}(\nu) = \ell^{-r} + O_{A_0}((\log Q)^{-A_0})$$

for all sufficiently large Q .

5 The Bernoulli-extremal obstruction

The affine counterexample of [Proposition 1](#) kills deterministic finite-field smoothing. We record here that the same failure persists under *Bernoulli thinning* of the prime set, ruling out a class of extremal-only proof strategies for the Bernoulli branch.

Lemma 1 (Bernoulli shadow). *Let $V = \mathbb{F}_\ell^r$, let $\mathcal{H}$ be a finite family of affine or linear hyperplanes in V , and let $\{x_i\}$ be a finite multiset in V with independent Bernoulli weights $\xi_i \sim \text{Bernoulli}(p_i)$. Define*

$$M = \sum_i p_i, \quad X = \sum_i \xi_i, \quad \mu(H) = \sum_{x_i \in H} p_i, \quad X(H) = \sum_{x_i \in H} \xi_i.$$

For $\delta \in (0, 1)$, set $L = \log(2(|\mathcal{H}| + 1)/\delta)$ and $U = 2\sqrt{ML} + 2L$. Then with probability $\geq 1 - \delta$, simultaneously for every $H \in \mathcal{H}$,

$$|X - M| \leq U, \quad |X(H) - \mu(H)| \leq U.$$

If $\mu(H_0) \geq (\theta + \eta)M$ for some H_0 and $U \leq \eta M/6$, then $X(H_0)/X \geq \theta + \eta/2$.

Proof. Bernstein's inequality applied to $X(H) - \mu(H) = \sum_i \mathbf{1}[x_i \in H](\xi_i - p_i)$, then a union bound over $\mathcal{H} \cup \{V\}$. $\square$

Corollary 3 (Bernoulli does not repair hyperplane support). *If every x_i with $p_i > 0$ lies in a common hyperplane H_0 , then $X(H_0) = X$ identically. If $M \rightarrow \infty$, then $\mathbb{P}(X = 0) \leq \exp(-M) \rightarrow 0$, so the Bernoulli sample is, with probability $1 - o(1)$, nonempty and 100% hyperplane-concentrated.*

We exhibit an explicit arithmetic family where this happens.

Lemma 2 (Hyperplane prime family). *Fix an odd prime ℓ and $A = (A_1, A_2) \in \mathbb{F}_\ell^2 \setminus \{0\}$. Set*

$$\alpha_A = 2^{A_1} 3^{A_2}, \quad K_{\ell, \star} = \mathbb{Q}(\zeta_\ell, 2^{1/\ell}, 3^{1/\ell}).$$

By rank-two Kummer independence (companion Paper 6), $\text{Gal}(K_{\ell, \star}/\mathbb{Q}(\zeta_\ell)) \simeq \mathbb{F}_\ell^2$. The prime set

$$\mathcal{P}_{\ell, A} = \{q \nmid 6\ell : q \equiv 1 \pmod{\ell}, x_\ell(q) \in H_A\}, \quad H_A = \{v \in \mathbb{F}_\ell^2 : A \cdot v = 0\},$$

has natural density $\delta_{\ell, A} = 1/(\ell(\ell - 1))$ by qualitative Chebotarev for $K_{\ell, \star}/\mathbb{Q}$.

Proof. The hyperplane atom criterion (Paper 6) gives $x_\ell(q) \in H_A \Leftrightarrow \alpha_A \in (\mathbb{F}_q^*)^\ell$. The absolute Galois group of $K_{\ell, \star}/\mathbb{Q}$ has order $\ell^2(\ell - 1)$; the conditions $q \equiv 1 \pmod{\ell}$ (factor $1/(\ell - 1)$) and $x_\ell(q) \in H_A$ (factor $|H_A|/|V| = 1/\ell$) compose to density $1/(\ell(\ell - 1))$. $\square$

Theorem 3 (Intrinsic Bernoulli-Kummer hyperplane obstruction). *Fix an odd prime ℓ and nonzero $A \in \mathbb{F}_\ell^2$. For $X \geq 2$ let*

$$\mathcal{Q}_X = \mathcal{P}_{\ell, A} \cap [1, X], \quad P_X = \prod_{q \in \mathcal{Q}_X} q,$$

and choose m uniformly in $(\mathbb{Z}/P_X\mathbb{Z})^$. Define*

$$W_X = \sum_{q \in \mathcal{Q}_X} B_q(m), \quad W_X(H_A) = \sum_{\substack{q \in \mathcal{Q}_X \\ x_\ell(q) \in H_A}} B_q(m).$$

Then $W_X(H_A) = W_X$ identically, and $\mathbb{P}_m(W_X > 0) \rightarrow 1$ as $X \rightarrow \infty$.

Proof. By Lemma 2, every $q \in \mathcal{Q}_X$ satisfies $x_\ell(q) \in H_A$, so $W_X(H_A) = W_X$ identically. By Mertens applied to a density- $1/(\ell(\ell - 1))$ prime set, $M_X = \mathbb{E}_m W_X \geq \sum_{q \in \mathcal{Q}_X} 1/(q - 1) = (\ell(\ell - 1))^{-1} \log \log X + O(1) \rightarrow \infty$, hence $\mathbb{P}(W_X = 0) \leq \exp(-M_X) \rightarrow 0$. $\square$

Corollary 4 (Bernoulli-extremal bypass kill). *No proof of Bernoulli-weighted PHNC at level ℓ using only extremal arguments on the Kummer log vectors $\{x_\ell(q)\} \subset \mathbb{F}_\ell^2$ can avoid producing the deterministic p_q -weighted Kummer equidistribution estimate*

$$\max_{A \neq 0} \left| \sum_{\substack{q \leq X \\ q \equiv 1 \pmod{\ell}}} p_q \left(\mathbf{1}_{A \cdot x_\ell(q)=0} - \frac{1}{\ell} \right) \right| = o \left(\sum_{\substack{q \leq X \\ q \equiv 1 \pmod{\ell}}} p_q \right).$$

Proof. Suppose an extremal-only proof exists. Specialise to the prime set $\mathcal{Q}_X = \mathcal{P}_{\ell, A_0} \cap [1, X]$ for some chosen A_0 . By [Theorem 3](#), the Bernoulli sample is hyperplane-concentrated on H_{A_0} with probability $1 - o(1)$. Bernoulli-weighted PHNC restricted to $\mathcal{Q}_X$ would force this concentration to be incompatible with prime distribution, which is impossible unless the proof controls the deterministic hyperplane discrepancy on $\mathcal{Q}_X$ — the displayed quantity. $\square$

The takeaway is methodological: the Bernoulli branch of KRWS ([Theorem 2](#)) is the correct finite reduction, but the deterministic weighted Kummer equidistribution input it requires cannot be replaced by purely extremal $\mathbb{F}_\ell^2$ combinatorics, even after Bernoulli thinning.

6 KRWS status

Paper 4's KRWS input should be read in two branches.

1. **Bernoulli branch.** For the obstruction vector $V(m) = \sum_q B_q(m)x_q$, KRWS is the finite theorem [Theorem 2](#), conditional only on the same weighted PHNC input used in Paper 11. No affine hypothesis is needed, because every local step is anchored by the 0-increment option.
2. **Deterministic branch.** For the raw prime-vector step measure $\mu_{\mathcal{Q}} = |\mathcal{Q}|^{-1} \sum_q \delta_{x_q}$, linear hyperplane non-concentration is insufficient by [Proposition 1](#). The correct finite replacement is affine non-concentration, which gives [Theorem 1](#).

This closes the finite L^2 collision-smoothing part of the Bernoulli-Kummer route under PHNC. It does not close the analytic sieve-transfer part. KRC, RTHLS, Kummer-BV, and BVK still encode the passage from finite obstruction-law smoothing to estimates for the von Mangoldt-weighted sequence $m2^k3^\ell + 1$.

7 Dependency status

- The affine counterexample is unconditional and finite.
- The deterministic smoothing theorem is unconditional once affine non-concentration is assumed.
- The Bernoulli-Kummer smoothing theorem is unconditional once the weighted PHNC mass $W_\ell(\mathcal{Q})$ is supplied.
- The PHNC high- ℓ corollary is conditional on PHNC and a density floor for the local Bernoulli parameters.
- No unconditional proof of PHNC, Kummer-BV, BVK, or Erdős–Graham #203 is asserted.